

Yahoo Finance e Python : parte 1

In questo Notebook illustrerò le funzioni messe a disposizione dalla principale libreria python che fornisce accesso ai dati di Yahoo Finance.

Lavorerò con un singolo 'ticker' rimandando alla parte 2 l'illustrazione dell'analisi relativa a più 'tickers' in cui mostrerò come effettuare il download di dati più complessi e articolati (dati storici di più azioni, FX, tassi, indici, ETF..)

Principali riferimenti:

Per primo il sito del nostro data provider cioè Yahoo dove potete rendervi conto di quali dati sono disponibili quindi il link alla libreria Python

1. [yahoo finance](#)
2. [Yahoo Finance Python Library: yfinance](#) -

Il package yfinance è:

- user-friendly e versatile anche grazie all'integrazione con Pandas;
- supportata da un'ampia documentazione e dalla Community degli utenti e sviluppatori
- Attivamente mantenuta, con aggiornamenti e correzioni di bug.

D'altro canto potrebbero registrarsi a volte problemi di accuratezza dei dati come segnalato da alcuni utenti in relazione ad occasionali discrepanze o dati mancanti rispetto ai download diretti dal sito web di Yahoo Finance.

Quindi la libreria, seppure non possa considerarsi una scelta adatta per usi professionali, è semplice da interpretare e facile da utilizzare e dando accesso ad una buona quantità e varietà di dati e può essere sicuramente usata per sviluppare modelli e algoritmi

Se si hanno in mente obiettivi più "seri", cioè si vogliono mettere in produzione algoritmi di supporto al trading o supportare decisioni di investimento importanti e si ha a disposizione una postazione Bloomberg meglio utilizzare Bquant, ma questa è un'altra storia....

Il presente notebook può essere eseguito in [Google Colab](#) o in locale installando la libreria [yfinance](#)

```
In [ ]: !pip install yfinance
```

```
In [2]: #import necessary library
import pandas as pd
import numpy as np
import datetime
import yfinance as yf
```

```
In [42]: # Set display options for pandas dataframes
pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 1000)
pd.set_option('display.colheader_justify', 'center')
pd.set_option('display.precision', 3)
pd.options.display.float_format = '{:,.2f}'.format
```

```
In [43]: """
Scegliamo come esempio il ticker del Monte dei Paschi di Siena
Download delle informazioni principali
i dati sono restituiti in una variabile di tipo Dictionary
"""
TickerName = "BMPS.MI"
Stock = yf.Ticker(TickerName)
StockInfo = Stock.info
```

```
In [ ]: type(Stock)
```

```
In [45]: Stock.info.keys()
```

```
Out[45]: dict_keys(['address1', 'city', 'state', 'zip', 'country', 'phone', 'website', 'industry', 'industryKey', 'industryDisp', 'sector', 'sectorKey', 'sectorDisp', 'longBusinessSummary', 'fullTimeEmployees', 'companyOfficers', 'auditRisk', 'boardRisk', 'compensationRisk', 'shareHolderRightsRisk', 'overallRisk', 'governanceEpochDate', 'compensationAsOfEpochDate', 'maxAge', 'priceHint', 'previousClose', 'open', 'dayLow', 'dayHigh', 'regularMarketPreviousClose', 'regularMarketOpen', 'regularMarketDayLow', 'regularMarketDayHigh', 'dividendRate', 'dividendYield', 'exDividendDate', 'payoutRatio', 'fiveYearAvgDividendYield', 'beta', 'trailingPE', 'forwardPE', 'volume', 'regularMarketVolume', 'averageVolume', 'averageVolume10days', 'averageDailyVolume10Day', 'bid', 'ask', 'marketCap', 'fiftyTwoWeekLow', 'fiftyTwoWeekHigh', 'priceToSalesTrailing12Months', 'fiftyDayAverage', 'twoHundredDayAverage', 'trailingAnnualDividendRate', 'trailingAnnualDividendYield', 'currency', 'enterpriseValue', 'profitMargins', 'floatShares', 'sharesOutstanding', 'heldPercentInsiders', 'heldPercentInstitutions', 'impliedSharesOutstanding', 'bookValue', 'priceToBook', 'lastFiscalYearEnd', 'nextFiscalYearEnd', 'mostRecentQuarter', 'earningsQuarterlyGrowth', 'netIncomeToCommon', 'trailingEps', 'forwardEps', 'pegRatio', 'lastSplitFactor', 'lastSplitDate', 'enterpriseToRevenue', '52WeekChange', 'SandP52WeekChange', 'lastDividendValue', 'lastDividendDate', 'exchange', 'quoteType', 'symbol', 'underlyingSymbol', 'shortName', 'longName', 'firstTradeDateEpochUtc', 'timeZoneFullName', 'timeZoneShortName', 'uuid', 'messageBoardId', 'gmtOffsetMilliseconds', 'currentPrice', 'targetHighPrice', 'targetLowPrice', 'targetMeanPrice', 'targetMedianPrice', 'recommendationMean', 'recommendationKey', 'numberOfAnalystOpinions', 'totalCash', 'totalCashPerShare', 'totalDebt', 'totalRevenue', 'revenuePerShare', 'returnOnAssets', 'returnOnEquity', 'earningsGrowth', 'revenueGrowth', 'operatingMargins', 'financialCurrency', 'trailingPegRatio'])
```

```
In [46]: StockInfo
```



```
{'maxAge': 1,
  'name': 'Mr. Alessandro Giacometti',
  'title': 'Chief Operating Officer',
  'fiscalYear': 2022,
  'exercisedValue': 0,
  'unexercisedValue': 0},
{'maxAge': 1,
  'name': 'Mr. Giovanni Damiani',
  'title': 'Chief Information Office',
  'fiscalYear': 2022,
  'exercisedValue': 0,
  'unexercisedValue': 0},
{'maxAge': 1,
  'name': 'Ms. Elisabetta Pozzi',
  'title': 'Head of Investor Relations',
  'fiscalYear': 2022,
  'exercisedValue': 0,
  'unexercisedValue': 0},
{'maxAge': 1,
  'name': 'Mr. Renzo Filippo Riccardo Quagliana',
  'title': 'Group General Counsel',
  'fiscalYear': 2022,
  'exercisedValue': 0,
  'unexercisedValue': 0},
{'maxAge': 1,
  'name': 'Mr. Ettore Carneade',
  'title': 'Chief Compliance Executive',
  'fiscalYear': 2022,
  'exercisedValue': 0,
  'unexercisedValue': 0},
{'maxAge': 1,
  'name': 'Ms. Fiorella Ferri',
  'title': 'Chief Human Capital Officer',
  'fiscalYear': 2022,
  'exercisedValue': 0,
  'unexercisedValue': 0},
{'maxAge': 1,
  'name': 'Mr. Maurizio Bai',
  'title': 'Deputy Commercial General Manager',
  'fiscalYear': 2022,
  'exercisedValue': 0,
```

```
'unexercisedValue': 0},
{'maxAge': 1,
 'name': 'Mr. Fabrizio Leandri',
 'age': 57,
 'title': 'Chief Lending Officer',
 'yearBorn': 1966,
 'fiscalYear': 2022,
 'exercisedValue': 0,
 'unexercisedValue': 0}],
'auditRisk': 6,
'boardRisk': 1,
'compensationRisk': 4,
'shareHolderRightsRisk': 1,
'overallRisk': 1,
'governanceEpochDate': 1722470400,
'compensationAsOfEpochDate': 1672444800,
'maxAge': 86400,
'priceHint': 2,
'previousClose': 5.148,
'open': 5.176,
'dayLow': 5.092,
'dayHigh': 5.176,
'regularMarketPreviousClose': 5.148,
'regularMarketOpen': 5.176,
'regularMarketDayLow': 5.092,
'regularMarketDayHigh': 5.176,
'dividendRate': 0.25,
'dividendYield': 0.048699997,
'exDividendDate': 1716163200,
'payoutRatio': 0.1204,
'fiveYearAvgDividendYield': 5.05,
'beta': 1.068,
'trailingPE': 2.4673078,
'forwardPE': 5.9674416,
'volume': 10023041,
'regularMarketVolume': 10023041,
'averageVolume': 17869588,
'averageVolume10days': 21580064,
'averageDailyVolume10Day': 21580064,
'bid': 5.114,
'ask': 5.116,
```

'marketCap': 6464729088,
'fiftyTwoWeekLow': 2.232,
'fiftyTwoWeekHigh': 5.364,
'priceToSalesTrailing12Months': 1.833248,
'fiftyDayAverage': 4.80592,
'twoHundredDayAverage': 4.02326,
'trailingAnnualDividendRate': 0.25,
'trailingAnnualDividendYield': 0.04856255,
'currency': 'EUR',
'enterpriseValue': -1339106688,
'profitMargins': 0.73503,
'floatShares': 921174247,
'sharesOutstanding': 1259689984,
'heldPercentInsiders': 0.26733,
'heldPercentInstitutions': 0.22635001,
'impliedSharesOutstanding': 1265069952,
'bookValue': 8.586,
'priceToBook': 0.59771717,
'lastFiscalYearEnd': 1703980800,
'nextFiscalYearEnd': 1735603200,
'mostRecentQuarter': 1719705600,
'earningsQuarterlyGrowth': 1.156,
'netIncomeToCommon': 2591981056,
'trailingEps': 2.08,
'forwardEps': 0.86,
'pegRatio': -1.71,
'lastSplitFactor': '1:100',
'lastSplitDate': 1664150400,
'enterpriseToRevenue': -0.38,
'52WeekChange': 0.89372694,
'SandP52WeekChange': 0.24822903,
'lastDividendValue': 0.25,
'lastDividendDate': 1716163200,
'exchange': 'MIL',
'quoteType': 'EQUITY',
'symbol': 'BMPS.MI',
'underlyingSymbol': 'BMPS.MI',
'shortName': 'BANCA MONTE PASCHI SIENA',
'longName': 'Banca Monte dei Paschi di Siena S.p.A.',
'firstTradeDateEpochUtc': 930294000,
'timeZoneFullName': 'Europe/Rome',

```

'timeZoneShortName': 'CEST',
'uuid': '192c6fc0-fc94-39a2-8693-2c29dfc3d6c6',
'messageBoardId': 'finmb_691202',
'gmtOffSetMilliseconds': 7200000,
'currentPrice': 5.132,
'targetHighPrice': 6.6,
'targetLowPrice': 5.3,
'targetMeanPrice': 6.1,
'targetMedianPrice': 6.2,
'recommendationMean': 2.4,
'recommendationKey': 'buy',
'numberOfAnalystOpinions': 8,
'totalCash': 20600999936,
'totalCashPerShare': 16.386,
'totalDebt': 12809199616,
'totalRevenue': 3526380032,
'revenuePerShare': 2.803,
'returnOnAssets': 0.02078,
'returnOnEquity': 0.26864,
'earningsGrowth': 1.16,
'revenueGrowth': 0.008,
'operatingMargins': 0.50493,
'financialCurrency': 'EUR',
'trailingPegRatio': None}

```

In [47]:

```

"""
Download dei dati storici usando data inizio e fine
frequenza giornaliera, data inizio e fine possono essere fornite in input anche nel formato di stringhe
"""

tipoinput = 0

if tipoinput == 0 :
    start = datetime.datetime(2022,2,1)
    end = datetime.datetime(2024,6,30)
else:
    start = '2022-2-1'
    end = '2023-12-31'

StockHist = Stock.history(start= start, end= end)

type(StockHist)

```



```
StockHist['Volume'] = StockHist.apply(lambda x: "{:,.0f}".format(x['Volume']), axis=1)
StockHist.head()
```

Out[47]:

	Open	High	Low	Close	Volume	Dividends	Stock Splits
Date							
2022-02-01 00:00:00+01:00	18.24	18.50	17.99	18.29	148,931	0.00	0.00
2022-02-02 00:00:00+01:00	18.44	19.02	18.30	18.66	378,368	0.00	0.00
2022-02-03 00:00:00+01:00	19.00	19.00	18.57	18.83	232,939	0.00	0.00
2022-02-04 00:00:00+01:00	18.98	19.06	18.26	18.36	264,963	0.00	0.00
2022-02-07 00:00:00+01:00	18.36	18.72	18.26	18.46	194,072	0.00	0.00

In [8]:

```
"""
Download dei dati storici usando period (definisce la lunghezza dell'intervallo di tempo) e frequenza giornaliera
"""
StockHist = Stock.history(period='max')
type(StockHist)
StockHist['Volume'] = StockHist.apply(lambda x: "{:,.0f}".format(x['Volume']), axis=1)
StockHist.head()
```

Out[8]:

	Open	High	Low	Close	Volume	Dividends	Stock Splits
Date							
1999-06-25 00:00:00+02:00	263,174,716.80	265,651,652.77	254,505,419.30	262,555,472.00	1,982	0.00	0.00
1999-06-28 00:00:00+02:00	266,270,872.87	269,367,031.74	262,555,447.65	266,890,096.00	812	0.00	0.00
1999-06-29 00:00:00+02:00	268,747,845.77	268,747,845.77	260,697,792.70	263,793,952.00	455	0.00	0.00
1999-06-30 00:00:00+02:00	264,413,199.12	266,270,912.00	261,317,018.19	266,270,912.00	232	0.00	0.00
1999-07-01 00:00:00+02:00	267,509,360.16	278,036,348.62	265,032,402.63	272,463,232.00	393	0.00	0.00

Analisi delle principali informazioni ed eventi

```
In [10]: StockActions = Stock.actions  
StockActions.head()
```

```
Out[10]:
```

	Dividends	Stock Splits
--	-----------	--------------

Date		
2000-05-22 00:00:00+02:00	7,164,389.50	0.00
2001-05-21 00:00:00+02:00	8,955,487.00	0.00
2002-05-20 00:00:00+02:00	4,625.51	0.00
2003-05-19 00:00:00+02:00	3,725.48	0.00
2004-05-24 00:00:00+02:00	2,444.85	0.00

```
In [11]: StockDividends = Stock.dividends  
StockDividends.head()
```

```
Out[11]:
```

Dividends

Date	
2000-05-22 00:00:00+02:00	7,164,389.50
2001-05-21 00:00:00+02:00	8,955,487.00
2002-05-20 00:00:00+02:00	4,625.51
2003-05-19 00:00:00+02:00	3,725.48
2004-05-24 00:00:00+02:00	2,444.85

dtype: float64

Dati di Bilancio e Finanziari

```
In [ ]: #Dati di Bilancio  
        Stock.balance_sheet
```

Out[]:

	2023-12-31	2022-12-31	2021-12-31	2020-12-31	2019-12-31
Treasury Shares Number	NaN	NaN	36,280,748.00	1,741,168.00	1,741,168.00
Ordinary Shares Number	1,259,689,706.00	1,259,689,706.00	48,106,975.00	46,365,807.00	NaN
Share Issued	1,259,689,706.00	1,259,689,706.00	48,106,975.00	48,106,975.00	NaN
Net Debt	NaN	NaN	NaN	NaN	10,511,167,000.00
Total Debt	12,941,704,000.00	10,303,201,000.00	12,737,640,000.00	16,536,002,000.00	NaN
Tangible Book Value	9,800,234,000.00	7,697,345,000.00	5,987,423,000.00	5,587,659,000.00	NaN
Invested Capital	22,729,006,000.00	17,952,002,000.00	18,684,125,000.00	22,084,881,000.00	NaN
Net Tangible Assets	9,800,234,000.00	7,697,345,000.00	5,987,423,000.00	5,587,659,000.00	NaN
Common Stock Equity	9,978,458,000.00	7,859,994,000.00	6,172,652,000.00	5,771,604,000.00	NaN
Total Capitalization	22,729,006,000.00	17,952,002,000.00	18,684,125,000.00	22,084,881,000.00	NaN
Total Equity Gross Minority Interest	9,979,109,000.00	7,860,930,000.00	6,173,998,000.00	5,772,914,000.00	NaN
Minority Interest	651,000.00	936,000.00	1,346,000.00	1,310,000.00	NaN
Stockholders Equity	9,978,458,000.00	7,859,994,000.00	6,172,652,000.00	5,771,604,000.00	NaN
Fixed Assets Revaluation Reserve	NaN	NaN	NaN	NaN	66,394,000.00
Treasury Stock	NaN	NaN	0.00	313,710,000.00	313,710,000.00
Retained Earnings	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
Capital Stock	7,453,451,000.00	7,453,451,000.00	9,195,012,000.00	9,195,012,000.00	NaN
Common Stock	7,453,451,000.00	7,453,451,000.00	9,195,012,000.00	9,195,012,000.00	NaN
Total Liabilities Net Minority Interest	112,634,582,000.00	112,374,374,000.00	131,694,564,000.00	144,572,122,000.00	NaN
Derivative Product Liabilities	1,345,635,000.00	1,645,538,000.00	2,701,965,000.00	3,298,998,000.00	NaN
Long Term Debt And Capital Lease Obligation	12,941,704,000.00	10,303,201,000.00	12,737,640,000.00	16,536,002,000.00	NaN

	2023-12-31	2022-12-31	2021-12-31	2020-12-31	2019-12-31
Payables	1,157,305,000.00	1,333,541,000.00	566,996,000.00	707,113,000.00	NaN
Other Payable	740,324,000.00	967,711,000.00	154,164,000.00	284,580,000.00	NaN
Total Tax Payable	221,357,000.00	167,977,000.00	135,340,000.00	122,254,000.00	NaN
Accounts Payable	195,624,000.00	197,853,000.00	277,492,000.00	300,279,000.00	NaN
Total Assets	122,613,691,000.00	120,235,304,000.00	137,868,562,000.00	150,345,036,000.00	NaN
Investments And Advances	20,218,313,000.00	21,649,133,000.00	26,248,553,000.00	26,781,336,000.00	NaN
Investmentin Financial Assets	19,491,622,000.00	20,898,455,000.00	25,153,141,000.00	25,673,873,000.00	NaN
Available For Sale Securities	13,220,401,000.00	14,115,944,000.00	15,013,577,000.00	16,057,785,000.00	NaN
Financial Assets Designatedas Fair Value Through Profit or Loss Total	245,475,000.00	314,759,000.00	322,582,000.00	317,885,000.00	NaN
Trading Securities	3,810,574,000.00	3,962,917,000.00	6,790,936,000.00	5,247,146,000.00	NaN
Long Term Equity Investment	726,691,000.00	750,678,000.00	1,095,412,000.00	1,107,463,000.00	NaN
Investment Properties	285,247,000.00	303,689,000.00	254,240,000.00	247,464,000.00	NaN
Goodwill And Other Intangible Assets	178,224,000.00	162,649,000.00	185,229,000.00	183,945,000.00	NaN
Other Intangible Assets	170,324,000.00	154,749,000.00	177,329,000.00	176,045,000.00	NaN
Goodwill	7,900,000.00	7,900,000.00	7,900,000.00	7,900,000.00	NaN
Net PPE	1,943,452,000.00	2,072,237,000.00	2,235,891,000.00	2,088,348,000.00	NaN
Gross PPE	1,943,452,000.00	2,072,237,000.00	2,235,891,000.00	2,088,348,000.00	NaN
Other Properties	95,545,000.00	118,984,000.00	145,591,000.00	194,496,000.00	NaN
Machinery Furniture Equipment	132,490,000.00	136,776,000.00	142,470,000.00	145,418,000.00	NaN
Buildings And Improvements	831,869,000.00	852,635,000.00	884,566,000.00	1,114,060,000.00	NaN
Land And Improvements	883,548,000.00	963,842,000.00	1,063,264,000.00	634,374,000.00	NaN

	2023-12-31	2022-12-31	2021-12-31	2020-12-31	2019-12-31
Prepaid Assets	523,278,000.00	512,071,000.00	516,785,000.00	506,993,000.00	NaN
Receivables	5,811,000.00	33,564,000.00	10,734,000.00	20,132,000.00	NaN
Taxes Receivable	2,197,551,000.00	1,718,293,000.00	1,147,190,000.00	1,056,754,000.00	NaN
Cash And Cash Equivalents	15,579,350,000.00	13,598,205,000.00	23,046,584,000.00	31,390,260,000.00	NaN
Cash Equivalents	NaN	NaN	NaN	78,883,000.00	57,406,000.00
Cash Financial	708,220,000.00	695,933,000.00	723,876,000.00	751,985,000.00	NaN
Cash Cash Equivalents And Federal Funds Sold	22,839,923,000.00	17,687,703,000.00	28,985,791,000.00	40,902,934,000.00	NaN

In [13]: `Stock.quarterly_balance_sheet.head()`

Out[13]:

	2024-06-30	2024-03-31	2023-12-31	2023-09-30	2023-06-30	2023-03-31	2022-12-31
Ordinary Shares Number	1,259,689,706.00	1,259,689,706.00	1,259,689,706.00	1,259,689,706.00	1,259,689,706.00	NaN	NaN
Share Issued	1,259,689,706.00	1,259,689,706.00	1,259,689,706.00	1,259,689,706.00	1,259,689,706.00	NaN	NaN
Total Debt	10,341,500,000.00	9,514,000,000.00	12,941,704,000.00	9,919,700,000.00	9,551,900,000.00	NaN	NaN
Tangible Book Value	10,787,100,000.00	10,299,200,000.00	9,800,234,000.00	8,799,200,000.00	8,362,400,000.00	NaN	NaN
Invested Capital	21,136,500,000.00	19,821,100,000.00	22,729,006,000.00	18,726,800,000.00	18,051,400,000.00	NaN	NaN

In [14]: `#Dati Finanziari
Stock.financials`

Out[14]:

	2023-12-31	2022-12-31	2021-12-31	2020-12-31	2019-12-31
Tax Effect Of Unusual Items	-26,592,240.00	-23,335,200.00	-22,336,800.00	-7,392,240.00	NaN
Tax Rate For Calcs	0.24	0.24	0.24	0.24	NaN
Total Unusual Items	-110,801,000.00	-97,230,000.00	-93,070,000.00	-30,801,000.00	NaN
Total Unusual Items Excluding Goodwill	-110,801,000.00	-97,230,000.00	-93,070,000.00	-30,801,000.00	NaN
Net Income From Continuing Operation Net Minority Interest	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
Reconciled Depreciation	181,084,000.00	193,466,000.00	208,567,000.00	243,143,000.00	NaN
Net Interest Income	2,291,928,000.00	1,535,236,000.00	1,217,048,000.00	1,271,446,000.00	NaN
Interest Expense	2,072,241,000.00	614,485,000.00	680,342,000.00	703,663,000.00	NaN
Interest Income	4,364,169,000.00	2,149,721,000.00	1,897,390,000.00	1,975,109,000.00	NaN
Normalized Income	2,135,989,760.00	-104,466,200.00	380,240,200.00	-1,663,098,240.00	NaN
Net Income From Continuing And Discontinued Operation	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
Diluted Average Shares	1,259,689,706.00	209,865,416.00	47,322,061.00	52,421,063.00	NaN
Basic Average Shares	1,259,689,706.00	209,865,416.00	47,322,061.00	52,421,063.00	NaN
Diluted EPS	1.63	-0.85	6.54	-32.17	NaN
Basic EPS	1.63	-0.85	6.54	-32.17	NaN
Diluted NI Availto Com Stockholders	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
Net Income Common Stockholders	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
Otherunder Preferred Stock Dividend	NaN	2,700,000.00	3,600,000.00	4,300,000.00	NaN
Net Income	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
Minority Interests	156,000.00	148,000.00	176,000.00	129,000.00	NaN
Net Income Including Noncontrolling Interests	2,051,625,000.00	-178,509,000.00	309,331,000.00	-1,686,636,000.00	NaN

	2023-12-31	2022-12-31	2021-12-31	2020-12-31	2019-12-31
Net Income Discontinuous Operations	NaN	NaN	NaN	0.00	-18,060,000.00
Net Income Continuous Operations	2,051,625,000.00	-178,509,000.00	309,331,000.00	-1,686,636,000.00	NaN
Tax Provision	-345,117,000.00	-426,620,000.00	-51,167,000.00	339,407,000.00	NaN
Pretax Income	1,706,508,000.00	-605,129,000.00	258,164,000.00	-1,347,229,000.00	NaN
Special Income Charges	-111,228,000.00	-97,452,000.00	-110,771,000.00	-85,146,000.00	NaN
Other Special Charges	53,144,000.00	31,111,000.00	44,134,000.00	19,113,000.00	NaN
Write Off	NaN	NaN	NaN	1,262,000.00	-1,000.00
Impairment Of Capital Assets	718,000.00	705,000.00	3,277,000.00	15,072,000.00	NaN
Restructuring And Mergern Acquisition	57,366,000.00	66,341,000.00	66,637,000.00	66,033,000.00	NaN
Gain On Sale Of Security	8,844,000.00	41,946,000.00	-11,767,000.00	-2,065,000.00	NaN
Operating Expense	1,805,689,000.00	3,459,164,000.00	2,729,555,000.00	3,801,311,000.00	NaN
Other Operating Expenses	-201,889,000.00	244,992,000.00	402,368,000.00	1,288,355,000.00	NaN
Depreciation And Amortization In Income Statement	181,084,000.00	193,466,000.00	208,567,000.00	243,143,000.00	NaN
Amortization	66,587,000.00	68,586,000.00	67,451,000.00	75,429,000.00	NaN
Depreciation Income Statement	114,497,000.00	124,880,000.00	141,116,000.00	167,714,000.00	NaN
Selling General And Administration	523,907,000.00	590,118,000.00	586,274,000.00	659,080,000.00	NaN
Selling And Marketing Expense	6,316,000.00	3,139,000.00	3,459,000.00	5,448,000.00	NaN
General And Administrative Expense	517,591,000.00	586,979,000.00	582,815,000.00	653,632,000.00	NaN
Insurance And Claims	NaN	19,857,000.00	41,329,000.00	47,453,000.00	49,216,000.00
Rent And Landing Fees	163,215,000.00	170,009,000.00	162,712,000.00	162,515,000.00	NaN
Total Revenue	3,982,436,000.00	3,287,501,000.00	3,183,800,000.00	3,183,990,000.00	NaN
Operatina Revenue	3,982,436,000.00	3,287,501,000.00	3,183,800,000.00	3,183,990,000.00	NaN


```
In [16]: Stock.quarterly_financials.head()
```

Out[16]:

	2024-06-30	2024-03-31	2023-12-31	2023-09-30	2023-06-30	2023-03-31	2022-12-31
Tax Effect Of Unusual Items	-12,720,000.00	-80,160.62	-9,024,000.00	-3,144,000.00	-573,569.62	NaN	NaN
Tax Rate For Calcs	0.24	0.01	0.24	0.24	0.03	NaN	NaN
Total Unusual Items	-53,000,000.00	-7,700,000.00	-37,600,000.00	-13,100,000.00	-19,200,000.00	NaN	NaN
Total Unusual Items Excluding Goodwill	-53,000,000.00	-7,700,000.00	-37,600,000.00	-13,100,000.00	-19,200,000.00	NaN	NaN
Net Income From Continuing Operation Net Minority Interest	826,500,000.00	332,700,000.00	1,123,200,000.00	309,600,000.00	383,300,000.00	NaN	NaN

```
In [17]: Stock.cashflow
```

Out[17]:

	2023-12-31	2022-12-31	2021-12-31	2020-12-31	2019-12-31
Free Cash Flow	1,662,907,000.00	8,315,752,000.00	-662,857,000.00	-1,227,560,000.00	NaN
Capital Expenditure	-111,438,000.00	-65,710,000.00	-111,321,000.00	-145,443,000.00	NaN
End Cash Position	14,317,277,000.00	12,538,578,000.00	1,741,766,000.00	2,319,441,000.00	NaN
Other Cash Adjustment Outside Changein Cash	1,000.00	NaN	NaN	NaN	NaN
Beginning Cash Position	12,538,578,000.00	1,741,766,000.00	2,319,441,000.00	3,408,419,000.00	NaN
Changes In Cash	1,778,698,000.00	10,796,812,000.00	-577,675,000.00	-1,088,978,000.00	NaN
Financing Cash Flow	-2,726,000.00	2,371,263,000.00	42,928,000.00	-331,000.00	NaN
Cash Dividends Paid	-96,000.00	-86,000.00	-53,000.00	-331,000.00	NaN
Common Stock Dividend Paid	-96,000.00	-86,000.00	-53,000.00	-331,000.00	NaN
Net Common Stock Issuance	-2,630,000.00	2,371,349,000.00	42,981,000.00	0.00	NaN
Investing Cash Flow	7,079,000.00	44,087,000.00	-69,067,000.00	-6,530,000.00	NaN
Dividends Received Cfi	116,367,000.00	107,267,000.00	30,370,000.00	638,000.00	NaN
Net Business Purchase And Sale	NaN	0.00	1,021,000.00	0.00	43,508,000.00
Sale Of Business	NaN	0.00	1,021,000.00	0.00	43,508,000.00
Net Intangibles Purchase And Sale	-83,033,000.00	-46,468,000.00	-68,943,000.00	-83,291,000.00	NaN
Sale Of Intangibles	NaN	0.00	2,633,000.00	99,000.00	217,000.00
Purchase Of Intangibles	-83,033,000.00	-46,468,000.00	-71,576,000.00	-83,390,000.00	NaN
Net PPE Purchase And Sale	-26,255,000.00	-16,712,000.00	-31,515,000.00	76,123,000.00	NaN
Sale Of PPE	2,150,000.00	2,530,000.00	8,230,000.00	138,176,000.00	NaN
Purchase Of PPE	-28,405,000.00	-19,242,000.00	-39,745,000.00	-62,053,000.00	NaN
Operating Cash Flow	1,774,345,000.00	8,381,462,000.00	-551,536,000.00	-1,082,117,000.00	NaN
Change In Working Capital	-97,294,000.00	8,613,799,000.00	-1,371,602,000.00	-1,602,279,000.00	NaN

	2023-12-31	2022-12-31	2021-12-31	2020-12-31	2019-12-31
Change In Other Current Liabilities	-40,375,000.00	-1,194,571,000.00	-973,590,000.00	-331,126,000.00	NaN
Change In Other Current Assets	-655,591,000.00	-113,263,000.00	745,870,000.00	140,324,000.00	NaN
Other Non Cash Items	-88,278,000.00	-91,011,000.00	-136,015,000.00	-156,381,000.00	NaN
Provisionand Write Offof Assets	368,513,000.00	748,386,000.00	665,424,000.00	2,183,704,000.00	NaN
Deferred Tax	-345,117,000.00	-426,620,000.00	-51,167,000.00	339,407,000.00	NaN
Gain Loss On Investment Securities	-115,104,000.00	-284,583,000.00	32,493,000.00	-159,932,000.00	NaN
Net Income From Continuing Operations	2,051,625,000.00	-178,509,000.00	309,331,000.00	-1,686,636,000.00	NaN

```
In [18]: Stock.quarterly_cashflow
```

```
Out[18]: —
```

```
In [19]: Stock.get_income_stmt()
```

Out[19]:

	2023-12-31	2022-12-31	2021-12-31	2020-12-31	2019-12-31
TaxEffectOfUnusualItems	-26,592,240.00	-23,335,200.00	-22,336,800.00	-7,392,240.00	NaN
TaxRateForCalcs	0.24	0.24	0.24	0.24	NaN
TotalUnusualItems	-110,801,000.00	-97,230,000.00	-93,070,000.00	-30,801,000.00	NaN
TotalUnusualItemsExcludingGoodwill	-110,801,000.00	-97,230,000.00	-93,070,000.00	-30,801,000.00	NaN
NetIncomeFromContinuingOperationNetMinorityInterest	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
ReconciledDepreciation	181,084,000.00	193,466,000.00	208,567,000.00	243,143,000.00	NaN
NetInterestIncome	2,291,928,000.00	1,535,236,000.00	1,217,048,000.00	1,271,446,000.00	NaN
InterestExpense	2,072,241,000.00	614,485,000.00	680,342,000.00	703,663,000.00	NaN
InterestIncome	4,364,169,000.00	2,149,721,000.00	1,897,390,000.00	1,975,109,000.00	NaN
NormalizedIncome	2,135,989,760.00	-104,466,200.00	380,240,200.00	-1,663,098,240.00	NaN
NetIncomeFromContinuingAndDiscontinuedOperation	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
DilutedAverageShares	1,259,689,706.00	209,865,416.00	47,322,061.00	52,421,063.00	NaN
BasicAverageShares	1,259,689,706.00	209,865,416.00	47,322,061.00	52,421,063.00	NaN
DilutedEPS	1.63	-0.85	6.54	-32.17	NaN
BasicEPS	1.63	-0.85	6.54	-32.17	NaN
DilutedNI Availto ComStockholders	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
NetIncomeCommonStockholders	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
OtherunderPreferredStockDividend	NaN	2,700,000.00	3,600,000.00	4,300,000.00	NaN
NetIncome	2,051,781,000.00	-178,361,000.00	309,507,000.00	-1,686,507,000.00	NaN
MinorityInterests	156,000.00	148,000.00	176,000.00	129,000.00	NaN
NetIncomeIncludingNoncontrollingInterests	2,051,625,000.00	-178,509,000.00	309,331,000.00	-1,686,636,000.00	NaN
NetIncomeDiscontinuousOperations	NaN	NaN	NaN	0.00	-18,060,000.00

	2023-12-31	2022-12-31	2021-12-31	2020-12-31	2019-12-31
NetIncomeContinuousOperations	2,051,625,000.00	-178,509,000.00	309,331,000.00	-1,686,636,000.00	NaN
TaxProvision	-345,117,000.00	-426,620,000.00	-51,167,000.00	339,407,000.00	NaN
PretaxIncome	1,706,508,000.00	-605,129,000.00	258,164,000.00	-1,347,229,000.00	NaN
SpecialIncomeCharges	-111,228,000.00	-97,452,000.00	-110,771,000.00	-85,146,000.00	NaN
OtherSpecialCharges	53,144,000.00	31,111,000.00	44,134,000.00	19,113,000.00	NaN
WriteOff	NaN	NaN	NaN	1,262,000.00	-1,000.00
ImpairmentOfCapitalAssets	718,000.00	705,000.00	3,277,000.00	15,072,000.00	NaN
RestructuringAndMergernAcquisition	57,366,000.00	66,341,000.00	66,637,000.00	66,033,000.00	NaN
GainOnSaleOfSecurity	8,844,000.00	41,946,000.00	-11,767,000.00	-2,065,000.00	NaN
OperatingExpense	1,805,689,000.00	3,459,164,000.00	2,729,555,000.00	3,801,311,000.00	NaN
OtherOperatingExpenses	-201,889,000.00	244,992,000.00	402,368,000.00	1,288,355,000.00	NaN
DepreciationAndAmortizationInIncomeStatement	181,084,000.00	193,466,000.00	208,567,000.00	243,143,000.00	NaN
Amortization	66,587,000.00	68,586,000.00	67,451,000.00	75,429,000.00	NaN
DepreciationIncomeStatement	114,497,000.00	124,880,000.00	141,116,000.00	167,714,000.00	NaN
SellingGeneralAndAdministration	523,907,000.00	590,118,000.00	586,274,000.00	659,080,000.00	NaN
SellingAndMarketingExpense	6,316,000.00	3,139,000.00	3,459,000.00	5,448,000.00	NaN
GeneralAndAdministrativeExpense	517,591,000.00	586,979,000.00	582,815,000.00	653,632,000.00	NaN
InsuranceAndClaims	NaN	19,857,000.00	41,329,000.00	47,453,000.00	49,216,000.00
RentAndLandingFees	163,215,000.00	170,009,000.00	162,712,000.00	162,515,000.00	NaN
TotalRevenue	3,982,436,000.00	3,287,501,000.00	3,183,800,000.00	3,183,990,000.00	NaN
OperatingRevenue	3,982,436,000.00	3,287,501,000.00	3,183,800,000.00	3,183,990,000.00	NaN

```
In [20]: Stock.get_income_stmt(freq='quarterly')
```

Out[20]:

	2024-06-30	2024-03-31	2023-12-31	2023-09-30	2023-06-30
TaxEffectOfUnusualItems	-12,720,000.00	-80,160.62	-9,024,000.00	-3,144,000.00	-573,569.62
TaxRateForCalcs	0.24	0.01	0.24	0.24	0.03
TotalUnusualItems	-53,000,000.00	-7,700,000.00	-37,600,000.00	-13,100,000.00	-19,200,000.00
TotalUnusualItemsExcludingGoodwill	-53,000,000.00	-7,700,000.00	-37,600,000.00	-13,100,000.00	-19,200,000.00
NetIncomeFromContinuingOperationNetMinorityInterest	826,500,000.00	332,700,000.00	1,123,200,000.00	309,600,000.00	383,300,000.00
ReconciledDepreciation	42,000,000.00	42,400,000.00	44,400,000.00	44,800,000.00	43,000,000.00
NetInterestIncome	585,200,000.00	587,000,000.00	604,200,000.00	605,000,000.00	578,300,000.00
InterestIncome	585,200,000.00	587,000,000.00	604,200,000.00	605,000,000.00	578,300,000.00
NormalizedIncome	866,780,000.00	340,319,839.38	1,151,776,000.00	319,556,000.00	401,926,430.38
NetIncomeFromContinuingAndDiscontinuedOperation	826,500,000.00	332,700,000.00	1,123,200,000.00	309,600,000.00	383,300,000.00
DilutedAverageShares	1,259,689,706.00	1,260,227,273.00	1,259,689,706.00	1,259,689,706.00	1,259,689,706.00
BasicAverageShares	1,259,689,706.00	1,260,227,273.00	1,259,689,706.00	1,259,689,706.00	1,259,689,706.00
DilutedEPS	0.66	0.26	0.89	0.25	0.30
BasicEPS	0.66	0.26	0.89	0.25	0.30
DilutedNI Availto ComStockholders	826,500,000.00	332,700,000.00	1,123,200,000.00	309,600,000.00	383,300,000.00
NetIncomeCommonStockholders	826,500,000.00	332,700,000.00	1,123,200,000.00	309,600,000.00	383,300,000.00
OtherunderPreferredStockDividend	NaN	NaN	NaN	NaN	NaN
NetIncome	826,500,000.00	332,700,000.00	1,123,200,000.00	309,600,000.00	383,300,000.00
MinorityInterests	100,000.00	0.00	100,000.00	0.00	100,000.00
NetIncomeIncludingNoncontrollingInterests	826,400,000.00	332,700,000.00	1,123,100,000.00	309,600,000.00	383,200,000.00
NetIncomeContinuousOperations	826,400,000.00	332,700,000.00	1,123,100,000.00	309,600,000.00	383,200,000.00
TaxProvision	-456,800,000.00	3,500,000.00	-338,800,000.00	-2,700,000.00	11,800,000.00

	2024-06-30	2024-03-31	2023-12-31	2023-09-30	2023-06-30
PretaxIncome	369,600,000.00	336,200,000.00	784,300,000.00	306,900,000.00	395,000,000.00
SpecialIncomeCharges	-53,000,000.00	-7,700,000.00	-37,600,000.00	-13,100,000.00	-19,200,000.00
OtherSpecialCharges	NaN	NaN	24,300,000.00	NaN	28,900,000.00
ImpairmentOfCapitalAssets	19,300,000.00	NaN	NaN	NaN	28,900,000.00
RestructuringAndMergernAcquisition	33,700,000.00	7,700,000.00	13,300,000.00	13,100,000.00	-9,700,000.00
GainOnSaleOfSecurity	3,800,000.00	-17,700,000.00	NaN	1,500,000.00	200,000.00
OperatingExpense	489,400,000.00	556,400,000.00	34,500,000.00	527,300,000.00	461,000,000.00
OtherOperatingExpenses	80,200,000.00	145,600,000.00	-450,500,000.00	132,700,000.00	11,800,000.00
DepreciationAndAmortizationInIncomeStatement	42,000,000.00	42,400,000.00	44,400,000.00	44,800,000.00	43,000,000.00
Amortization	16,600,000.00	16,700,000.00	NaN	17,400,000.00	NaN
DepreciationIncomeStatement	25,400,000.00	25,700,000.00	44,400,000.00	27,400,000.00	43,000,000.00
SellingGeneralAndAdministration	62,100,000.00	62,700,000.00	119,700,000.00	64,300,000.00	119,500,000.00
SellingAndMarketingExpense	600,000.00	500,000.00	NaN	700,000.00	NaN
GeneralAndAdministrativeExpense	61,500,000.00	62,200,000.00	119,700,000.00	63,600,000.00	119,500,000.00
InsuranceAndClaims	NaN	NaN	NaN	NaN	NaN
TotalRevenue	1,014,200,000.00	1,005,800,000.00	989,600,000.00	951,700,000.00	972,600,000.00
OperatingRevenue	1,014,200,000.00	1,005,800,000.00	989,600,000.00	951,700,000.00	972,600,000.00

```
In [22]: #Principali Azionisti
SH=Stock.get_major_holders()
SH
```


Out[22]:

Breakdown	Value
insidersPercentHeld	0.27
institutionsPercentHeld	0.23
institutionsFloatPercentHeld	0.31
institutionsCount	141.00

In [29]: *#Azionisti Istituzionali*
Stock.get_institutional_holders()

Out[29]: —

In []: *#Sustainability*
Stock.get_sustainability()

In [26]: *#Raccomandazioni degli analisti*
Stock.recommendations.sort_index(ascending=False)

Out[26]:

	period	strongBuy	buy	hold	sell	strongSell
3	-3m	0	0	1	2	0
2	-2m	0	0	1	2	0
1	-1m	0	0	2	3	0
0	0m	0	0	0	0	0

In [27]: *#Calendario Principali eventi*
Stock.calendar

```
Out[27]: {'Ex-Dividend Date': datetime.date(2024, 5, 20),
'Earnings Date': [datetime.date(2024, 11, 7)],
'Earnings High': 0.21,
'Earnings Low': 0.21,
'Earnings Average': 0.21,
'Revenue High': 1001000000,
'Revenue Low': 962000000,
'Revenue Average': 979800000}
```

```
In [28]: # show all earnings dates
Stock.earnings_dates
```

Out[28]:

	EPS Estimate	Reported EPS	Surprise(%)
Earnings Date			
2025-08-04 12:00:00+02:00	NaN	NaN	NaN
2025-05-05 12:00:00+02:00	NaN	NaN	NaN
2025-02-04 11:00:00+01:00	NaN	NaN	NaN
2024-11-07 11:00:00+01:00	NaN	NaN	NaN
2024-08-06 08:00:00+02:00	0.22	0.66	2.05
2024-08-05 12:00:00+02:00	0.21	NaN	NaN
2024-05-07 07:00:00+02:00	0.16	0.26	0.70
2024-05-06 12:00:00+02:00	0.15	0.26	0.73
2024-02-07 07:00:00+01:00	0.21	0.90	3.22
2023-11-08 08:00:00+01:00	0.15	0.24	0.60
2023-08-04 07:00:00+02:00	0.14	0.30	1.17
2023-05-13 01:00:00+02:00	0.13	0.19	0.44

```
In [30]: #Options expirations ()
```

```
Stock.options
```

```
Out[30]: ()
```

Grafici e visualizzazione

```
In [31]: StockHist = Stock.history(period='6mo')
StockHist.head()
```

```
Out[31]:
```

	Open	High	Low	Close	Volume	Dividends	Stock Splits
Date							
2024-02-14 00:00:00+01:00	3.38	3.49	3.34	3.48	26692588	0.00	0.00
2024-02-15 00:00:00+01:00	3.49	3.56	3.40	3.40	27477458	0.00	0.00
2024-02-16 00:00:00+01:00	3.45	3.45	3.34	3.36	14759944	0.00	0.00
2024-02-19 00:00:00+01:00	3.36	3.39	3.31	3.32	16175878	0.00	0.00
2024-02-20 00:00:00+01:00	3.32	3.34	3.29	3.29	11320910	0.00	0.00

```
In [32]: import plotly.graph_objects as go
fig = go.Figure(data=[go.Candlestick(x=StockHist.index.values, open=StockHist['Open'],
high=StockHist['High'], low=StockHist['Low'], close=StockHist['Close'])])
fig.update_layout(autosize=False, width=1400, height=800)
fig.show()
```



```
In [33]: import plotly.express as px
StockHist['DT']=StockHist.index.values
fig = px.line(StockHist, x="DT", y="Open", title='Stock Prices')
fig.show()
```

/usr/local/lib/python3.10/dist-packages/_plotly_utils/basevalidators.py:105: FutureWarning:

The behavior of DatetimeProperties.to_pydatetime is deprecated, in a future version this will return a Series containing python datetime objects instead of an ndarray. To retain the old behavior, call `np.array` on the result

```
In [34]: fig = go.Figure(data=go.Ohlc(x=StockHist['DT'], open=StockHist['Open'], high=StockHist['High'], low=StockHist['Low'], close=StockHist['Close']),
fig.update_layout(autosize=False, width=1400, height=900)
fig.show()
```

```
/usr/local/lib/python3.10/dist-packages/_plotly_utils/basevalidators.py:105: FutureWarning:
```

The behavior of `DatetimeProperties.to_pydatetime` is deprecated, in a future version this will return a Series containing python datetime objects instead of an ndarray. To retain the old behavior, call ``np.array`` on the result

